

Hot water into a cold pipe

The physics of dead-leg warm-up, and how the deadleg model represents it

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When a hot tap is opened at the end of a pipe that has gone cold, the first water out is the cold water that was standing in the pipe. That much is simple displacement. But the hot water that follows has to heat the pipe wall as it goes, and it loses a little heat to the air. So the tap runs warm later, and more water is wasted, than displacement alone suggests. This note sets out the physics of that transient: the governing equations, the few dimensionless numbers that control it, the analytical limits, and why copper and plastic pipes behave so differently. The model built on it predicts the delivery times of 46 field tests to within a few seconds, with no fitted constants.

The question

A pipe of length L and bore radius r_i holds a volume $V = \pi r_i^2 L$ of water. If hot water simply pushed the cold water out, like a piston, the hot water would reach the tap after the *plug-flow time*

$$t_{\text{plug}} = \frac{V}{Q} = \frac{L}{u}, \quad u = \frac{Q}{\pi r_i^2}, \quad (1)$$

where Q is the volume flow and u the mean velocity.¹ Everything else in this note explains why the real delivery time is longer than this, and by how much.

The setting is a domestic hot water branch: flows of 1 to 6 l/min in pipes of roughly 10 to 12 mm outside diameter, made of copper or plastic (PE-X). At the start, the water in the pipe, the pipe wall and the surrounding air are all at the same temperature T_0 , typically about 20 °C. At $t = 0$ the tap opens and water arrives at the inlet at T_{in} , for example 55 °C.

Four bodies, three exchanges

The model follows four bodies (Figure 1):

1. **the water**, flowing along the pipe as a plug, with a bulk temperature $T_w(x, t)$ that varies along the pipe;
2. **the pipe wall**, homogeneous and axially symmetric, with a temperature $T_p(r, x, t)$ that varies through the thickness and along the length;
3. **the air**, a single well-mixed body at $T_a(t)$ that warms as the pipe gives it heat;
4. optionally, **an environment** at a fixed temperature, to which the air body can lose heat.

¹ For 5 m of 10 mm copper (bore 8.8 mm), $V = 0.301$ and $t_{\text{plug}} = 6$ s at 3 l/min. At 1 l/min it is 18 s.

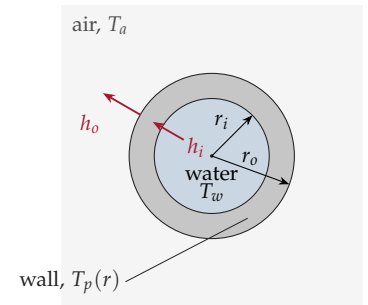


Figure 1: Cross-section. Heat passes from the water to the wall (film coefficient h_i), conducts through the wall, and leaves its outer surface for the air (h_o).

They exchange heat in a chain: water to wall by forced convection, through the wall by conduction, and wall to air by natural convection and radiation.

Governing equations

Water. An energy balance on a thin slice of water dx , carried along with the flow, gives

$$\rho_w c_w A_i \left(\frac{\partial T_w}{\partial t} + u \frac{\partial T_w}{\partial x} \right) = h_i 2\pi r_i (T_p(r_i) - T_w), \quad (2)$$

with $A_i = \pi r_i^2$, the inlet condition $T_w(0, t) = T_{in}(t)$ and the initial condition $T_w(x, 0) = T_0$. The left side is the rate of change of the heat held by the slice, seen by an observer moving with the water. The right side is the heat it gives to the wall.²

² Axial conduction in the water is neglected. The Péclet number uL/α_w is about 10^7 , so heat is carried along the pipe by the flow and hardly at all by conduction.

Wall. Transient conduction in an axially symmetric annulus,

$$\rho_p c_p \frac{\partial T_p}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(k_p r \frac{\partial T_p}{\partial r} \right) + k_p \frac{\partial^2 T_p}{\partial x^2}, \quad r_i < r < r_o, \quad (3)$$

with convective conditions on both faces,

$$-k_p \frac{\partial T_p}{\partial r} \Big|_{r_i} = h_i (T_w - T_p(r_i)), \quad (4)$$

$$-k_p \frac{\partial T_p}{\partial r} \Big|_{r_o} = h_o (T_p(r_o) - T_a). \quad (5)$$

The wall starts at T_0 , and its ends are taken as insulated.

Air. The air body has a single temperature, so its balance is an ordinary differential equation:

$$\rho_a c_a V_a \frac{dT_a}{dt} = \int_0^L h_o 2\pi r_o (T_p(r_o) - T_a) dx - UA_{env} (T_a - T_{env}). \quad (6)$$

Letting $V_a \rightarrow \infty$ holds the air at T_0 , which represents a large room.

Conservation. Adding the three equations and integrating over time gives the energy budget the numerical solution must respect:

$$\int_0^t \rho_w c_w Q (T_{in} - T_{out}) dt' = \Delta E_{\text{water}} + \Delta E_{\text{wall}} + \Delta E_{\text{air}} + \int_0^t UA_{env} (T_a - T_{env}) dt'. \quad (7)$$

In words, the heat that entered and did not leave at the tap is now in the water, the wall or the air, or has gone to the environment.

What controls the delay

Scaling equations (2)–(6) shows that three groups do almost all the work.

The heat capacity ratio.

$$\sigma = \frac{\rho_p c_p A_p}{\rho_w c_w A_i} \quad (8)$$

is the heat the wall can absorb per metre for each degree it warms, relative to the water's. Suppose the wall took up heat instantly, with no losses. Each metre of pipe would then have to absorb $(1 + \sigma)$ times the heat the water alone holds before it could pass hot water on. The hot front would therefore move at a speed of $u/(1 + \sigma)$ and arrive at

$$t_{\text{front}} = (1 + \sigma) t_{\text{plug}}. \quad (9)$$

This is the most useful single result in this note. Copper's thin wall gives $\sigma \approx 0.2$; plastic's thick wall gives $\sigma \approx 0.5$ to 0.6 (Table 1).³

The number of transfer units.

$$NTU = \frac{h_i 2\pi r_i L}{\rho_w c_w Q} \quad (10)$$

measures how completely the water hands its heat to the wall in one pass. If it were very large, the water and the inner surface of the wall would sit at the same temperature everywhere. Over 5 m of these pipes it is between about 1 and 4. That is large enough for the wall to take most of the heat from the leading water, and small enough that some lukewarm water arrives at the plug-flow time, ahead of the main rise. Above about 2 l/min, NTU hardly changes with flow, because h_i rises almost in proportion to the velocity.

The wall Biot number and diffusion time.

$$Bi = \frac{h_i \delta}{k_p}, \quad t_{\text{wall}} = \frac{\delta^2}{\alpha_p}, \quad \delta = r_o - r_i, \quad \alpha_p = \frac{k_p}{\rho_p c_p} \quad (11)$$

compare the resistance of the wall to the resistance of the water film, and give the time heat takes to soak through the wall. This is where copper and plastic part company:

- copper: $Bi \sim 0.01$, $t_{\text{wall}} \approx 4$ ms. The wall is isothermal through its thickness from the instant the water touches it, so it behaves as a simple lumped heat store;
- PE-X: $Bi \sim 10$ to 60 , $t_{\text{wall}} \approx 12$ to 20 s. The wall is a poor conductor; while the hot water passes, only a thin inner skin warms.

	bore	σ	t_{wall}	NTU / Bi at		
	mm		s	1 l/min	3 l/min	6 l/min
Copper 10×0.6	8.8	0.24	0.004	2.1 / 0.002	3.4 / 0.009	3.2 / 0.02
Copper 12×0.6	10.8	0.19	0.004	1.3 / 0.001	2.8 / 0.006	2.7 / 0.01
PE-X 10×1.5	7.0	0.50	11.7	3.2 / 8	4.2 / 31	3.9 / 57
PE-X 12×2.0	8.0	0.59	20.8	2.5 / 7	3.7 / 33	3.4 / 60

Figure 2 shows the consequence. Copper soaks up its share of heat at once, so the outlet stays cool until the delayed front of equation (9)

³ Per unit volume, copper and PE-X store heat in a ratio of only about 1.7 to 1. The difference in σ comes from the walls: a plastic wall is two to three times as thick.

Table 1: The controlling groups for a 5 m run, with water properties at 40 °C. NTU is quoted for the whole 5 m. Copper wall: $k = 340$ W/m K, $\rho c = 3.4$ MJ/m³K. PE-X: $k = 0.38$ W/m K, $\rho c = 2.0$ MJ/m³K.

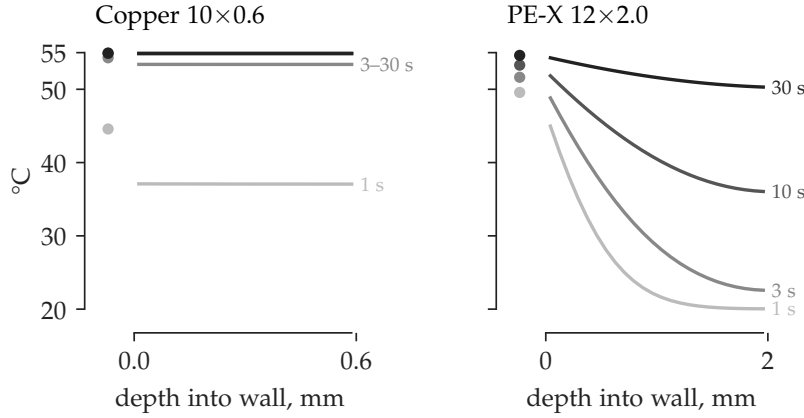


Figure 2: Temperature through the wall, half-way along a 5m run at 3 l/min, at 1, 3, 10 and 30 s after the hot front passes. The dots at the left are the water. Copper is uniform through its thickness and catches up with the water in a few seconds. In PE-X a steep gradient persists: after 10 s the outer half of the wall is still close to its starting temperature.

arrives, then rises steeply. PE-X takes only a skin of heat at first. Its effective σ is small early on and grows as heat penetrates the wall. So the first warm water arrives close to the plug-flow time, but the last few degrees wait for the wall to soak through.

Heat transfer coefficients

Inside. The film coefficient comes from the Reynolds number $Re = uD_i/\nu$. Above $Re = 10^4$ the model uses Gnielinski's correlation,

$$Nu = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f/8}(Pr^{2/3} - 1)}, \quad f = (0.790 \ln Re - 1.64)^{-2}, \quad (12)$$

and below $Re = 2300$ the fully developed laminar value $Nu = 3.66$. Between the two it interpolates linearly in Re , as the VDI Heat Atlas recommends.⁴ This gives h_i of about 10^3 to 1.5×10^4 W/m²K.

Outside. Natural convection from a horizontal cylinder (Churchill and Chu) plus radiation to surroundings at the air temperature, $h_{rad} = \epsilon\sigma_{SB}(T_s^2 + T_a^2)(T_s + T_a)$. For a 50 °C pipe in 20 °C air, this gives $h_o \approx 9$ W/m²K for tarnished copper ($\epsilon = 0.1$) and ≈ 14 W/m²K for plastic ($\epsilon = 0.9$).

The outside coefficient is a thousand times smaller than the inside one. While the tap is running, the pipe loses only a few watts per metre to the air. Over a draw of a minute or two, almost all of the heat leaving the water goes into the wall. External losses set the steady outlet temperature and how quickly the pipe cools between draws, but they barely change the delivery time. The model confirms this. Enclosing the pipe in a 1.5 m × 0.3 m air body instead of an infinite room changes no field-trial prediction by more than 0.34 s, even though the air warms by up to 7 K.

Exact solutions in limiting cases

Three limits have closed-form solutions. Each one checks the model and sharpens intuition.

⁴ At 1–6 l/min in 7–11 mm bores, Re is about 3000–28000, so the flow is transitional at the lowest flows and turbulent otherwise. Viscosity halves between 20 and 55 °C, so the arriving hot water is more turbulent than the cold water it displaces. The model evaluates properties at the local water temperature.

No wall heat capacity. The outlet steps from T_0 to T_{in} at t_{plug} , equation (1).

A thin, perfectly conducting wall with no losses. This is the Anzelius–Schumann problem from regenerator theory. In terms of

$$\xi = \frac{h_i P x}{\rho_w c_w Q}, \quad \eta = \frac{h_i P (t - x/u)}{\rho_p c_p A_p}, \quad P = 2\pi r_i,$$

the dimensionless water temperature $\theta = (T_w - T_0)/(T_{in} - T_0)$ is, for $\eta \geq 0$,

$$\theta = e^{-\xi} \left[e^{-\eta} I_0(2\sqrt{\xi\eta}) + \int_0^\eta e^{-s} I_0(2\sqrt{\xi s}) ds \right], \quad (13)$$

and zero before the plug front arrives. At $x = L$, ξ is the NTU . The factor $e^{-\xi}$ is the step of lukewarm water that arrives at the plug-flow time; the rest of the rise is spread around t_{front} of equation (9).

Steady state with losses. Once the wall has warmed, the outlet settles at

$$T_{out} = T_a + (T_{in} - T_a) \exp\left(-\frac{U'L}{\rho_w c_w Q}\right), \quad (14)$$

$$\frac{1}{U'} = \frac{1}{2\pi r_i h_i} + \frac{\ln(r_o/r_i)}{2\pi k_p} + \frac{1}{2\pi r_o h_o}. \quad (15)$$

Because h_o is small, the drop is a fraction of a degree over 5 m.

How the equations are solved

The pipe is divided into N_x cells along its length (200 by default). Each cell has one water node and N_r concentric wall shells, and there is a single air node. Each time step does three things:

1. **Radial heat exchange** between the water, the wall shells and the air, solved implicitly. Each cell's water-and-wall system is tridiagonal, and its solution is linear in the new air temperature. That lets the air node, which couples every cell, be eliminated exactly, so no iteration is needed;
2. **Axial conduction** along the wall, also implicit;
3. **Advection**: the time step is chosen as $\Delta t = \Delta x/u$, so the water moves exactly one cell per step. Transport is exact, and the front is not artificially smeared.

The scheme conserves energy to rounding error. It converges to the Anzelius–Schumann solution at first order: within 0.7% at 200 cells and 0.2% at 800 (Figure 3).

What the model predicts

Figure 4 shows both behaviours. Each copper trace is flat until t_{plug} , has a small step of lukewarm water, then rises steadily to about

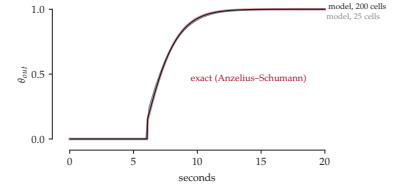


Figure 3: The numerical model (grey and black) against the exact Anzelius–Schumann solution (red), copper-like wall, 5 m at 3 l/min. Even at 25 cells the curves are almost indistinguishable.

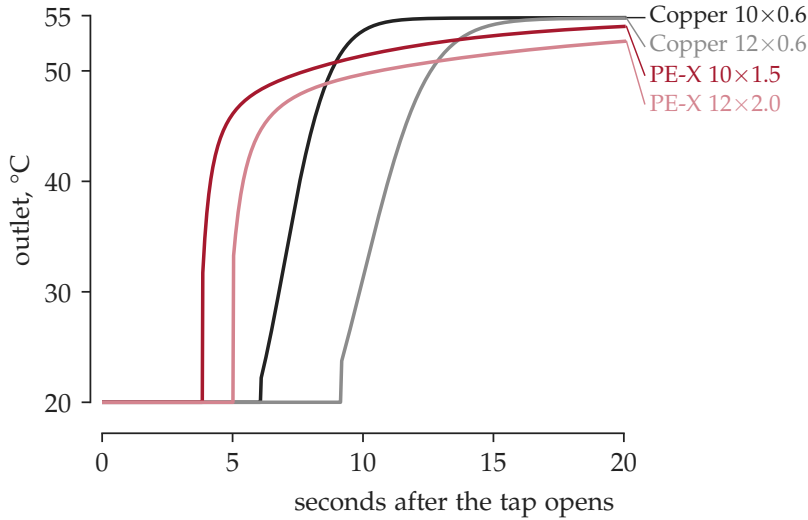


Figure 4: Outlet temperature after the tap opens: 5 m run, 3 l/min, inlet 55 °C, pipe and air initially 20 °C.

$(1 + \sigma)t_{\text{plug}}$ and beyond. The PE-X traces jump almost to 40 °C at the plug-flow time, then creep towards the inlet temperature as heat soaks into the wall.

Three practical consequences follow.

- **For copper, waste is fixed in volume.** About 1.4 to 1.5 pipe volumes run off before the tap reaches 50 °C, whatever the flow. The time therefore scales as $1/Q$.
- **For plastic, the answer depends on what “hot” means.** The time to 50 °C can be twice the time to 40 °C. At 1 l/min, a PE-X pipe is as slow to 50 °C as copper of the same outside diameter, despite holding less water. At 6 l/min, it is faster.
- **A delivery time is meaningful only with its threshold and supply temperature.** A 45 °C target with 50 °C supply sits in the slow tail for plastic pipe.

Checking against a field trial

Ridge and Jones (FairHeat, 2022) measured the time to 45 °C at the end of 5, 15 and 25 m runs of multilayer pipe at 4, 6 and 9 l/min, supplied by a heat interface unit, with the pipe starting at about 19 °C. The model was run for every test with no adjustment. The inlet stepped to 50 °C at the time the unit itself took to deliver hot water. The multilayer wall was represented as one homogenised material, and the air body was enclosed and 1.5 m × 0.3 m in section.

	model		paper’s regression	
	bias	RMS	bias	RMS
Manufacturer A (23 tests)	−0.1 s	3.3 s	+3.1 s	4.3 s
Manufacturer B, hold-out (23)	+1.1 s	7.0 s	+6.2 s	8.9 s

The physical model does better than the paper’s regression on

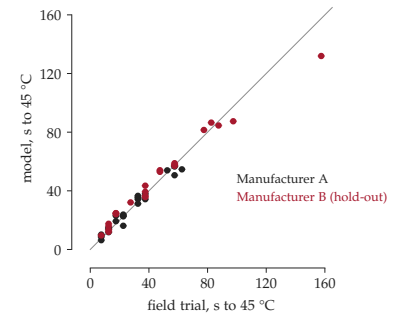


Figure 5: Model against measured time to 45 °C for all 46 field tests. The diagonal is perfect agreement. Each dot sits half-way through its 5 s reading window.

Table 2: Errors against the middle of each 5 s reading window. The paper’s regression was fitted to Manufacturer A.

both sets, including the set the regression was fitted to (Table 2, Figure 5). Its largest misses are the slow, long, large-bore runs at 4 l/min, where it predicts hot water 10 to 28 s early. The likeliest explanation, not yet tested, is that the heat unit supplied water a little below 50 °C at that flow. Because 45 °C is close to the supply temperature, a degree matters a great deal there.

Limitations

- **Plug flow.** Below $Re \approx 2300$ the parabolic velocity profile brings the first warm water out at about half the plug-flow time and smears the front. That matters only near or below 1 l/min with cold water.
- **Steady correlations.** A steady-state h_i is applied to a strongly transient wall. The entrance region and the moving thermal front both raise the true coefficient slightly.
- **A homogeneous wall.** Multilayer pipe is represented by volume-averaged properties.
- **What is left out.** Fittings, clips, insulation and the thermal mass of the surrounding structure are not modelled. Radiation is taken to go to surroundings at the air temperature.

The code, tests and data behind this note are in the GitHub repository `gwilma/deadleg_deadtime`. The file `docs/derivation.md` gives the same derivation in brief, and `docs/physics/make_figures.py` redraws every figure.